# Zomato Data Analysis

Understanding customer preferences and restaurant trends is important for making informed business decisions in food industry. In this article, we will analyze Zomato's restaurant dataset using Python to find meaningful insights.

We aim to answer questions such as: Do more restaurants provide online delivery compared to offline services? Which types of restaurants are most favored by the general public? What price range do couples prefer for dining out?

In [147…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [148…

```
dataframe = pd.read_csv("zomato-data.csv")
dataframe.head()
```

Out[148…

| | name | online_order | book_table | rate | votes | approx_cost(for two people) | listed_in(type) |
|---|---|---|---|---|---|---|---|
| **0** | Jalsa | Yes | Yes | 4.1/5 | 775 | 800 | Buffet |
| **1** | Spice Elephant | Yes | No | 4.1/5 | 787 | 800 | Buffet |
| **2** | San Churro Cafe | Yes | No | 3.8/5 | 918 | 800 | Buffet |
| **3** | Addhuri Udupi Bhojana | No | No | 3.7/5 | 88 | 300 | Buffet |
| **4** | Grand Village | No | No | 3.8/5 | 166 | 600 | Buffet |

In [149…

```
def handleRate(value):
    value=str(value).split('/')
    value=value[0];
    return float(value)

dataframe['rate']=dataframe['rate'].apply(handleRate)
dataframe.head()
```

Out[149…

| | name | online_order | book_table | rate | votes | approx_cost(for two people) | listed_in(type) |
|---|---|---|---|---|---|---|---|
| **0** | Jalsa | Yes | Yes | 4.1 | 775 | 800 | Buffet |
| **1** | Spice Elephant | Yes | No | 4.1 | 787 | 800 | Buffet |
| **2** | San Churro Cafe | Yes | No | 3.8 | 918 | 800 | Buffet |
| **3** | Addhuri Udupi Bhojana | No | No | 3.7 | 88 | 300 | Buffet |
| **4** | Grand Village | No | No | 3.8 | 166 | 600 | Buffet |

In [150…

```
dataframe.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   name                         148 non-null    object
 1   online_order                 148 non-null    object
 2   book_table                   148 non-null    object
 3   rate                         148 non-null    float64
 4   votes                        148 non-null    int64
 5   approx_cost(for two people)  148 non-null    int64
 6   listed_in(type)              148 non-null    object
dtypes: float64(1), int64(2), object(4)
memory usage: 8.2+ KB
```

In [151…

```
print(dataframe.isnull().sum())
```

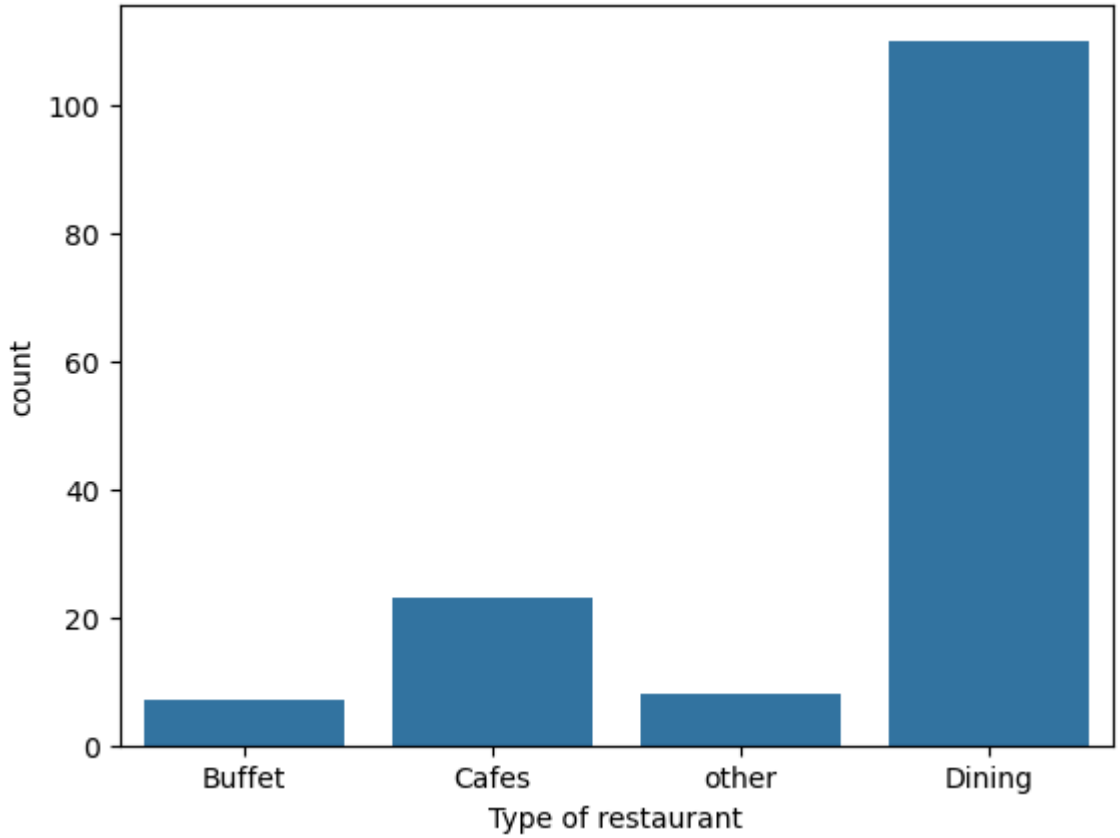
```
name                           0
online_order                   0
book_table                     0
rate                           0
votes                          0
approx_cost(for two people)    0
listed_in(type)                0
dtype: int64
```

In [152…

```
sns.countplot(x=dataframe['listed_in(type)'])
plt.xlabel("Type of restaurant")
```

Out[152…

```
Text(0.5, 0, 'Type of restaurant')
```




In [153…

```
grouped_data = dataframe.groupby('listed_in(type)')['votes'].sum()
result = pd.DataFrame({'votes': grouped_data})
plt.plot(result, c='green', marker='o')
plt.xlabel('Type of restaurant')
plt.ylabel('Votes')
```

Out[153…

```
Text(0, 0.5, 'Votes')
```

In [154…

```
max_votes = dataframe['votes'].max()
restaurant_with_max_votes = dataframe.loc[dataframe['votes'] == max_votes, 'name']

print('Restaurant(s) with the maximum votes:')
print(restaurant_with_max_votes)
```

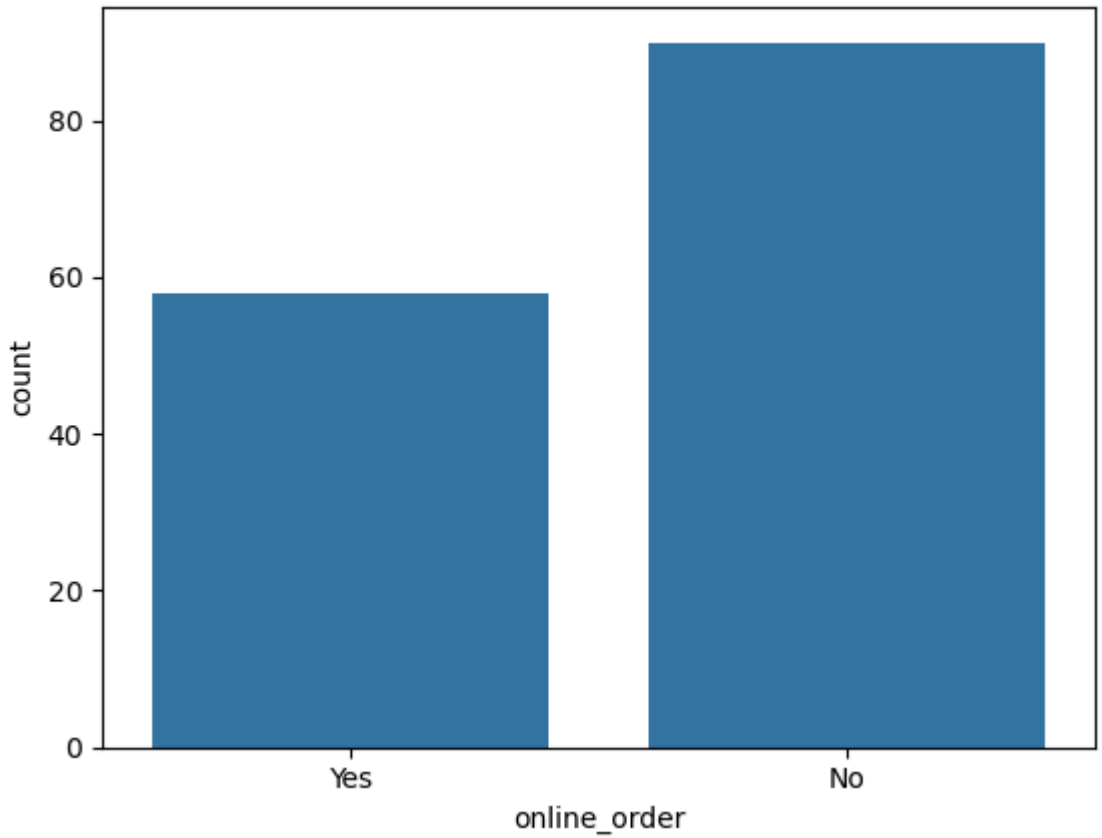
```
Restaurant(s) with the maximum votes:
38    Empire Restaurant
Name: name, dtype: object
```

In [155…

```
sns.countplot(x=dataframe['online_order'])
```

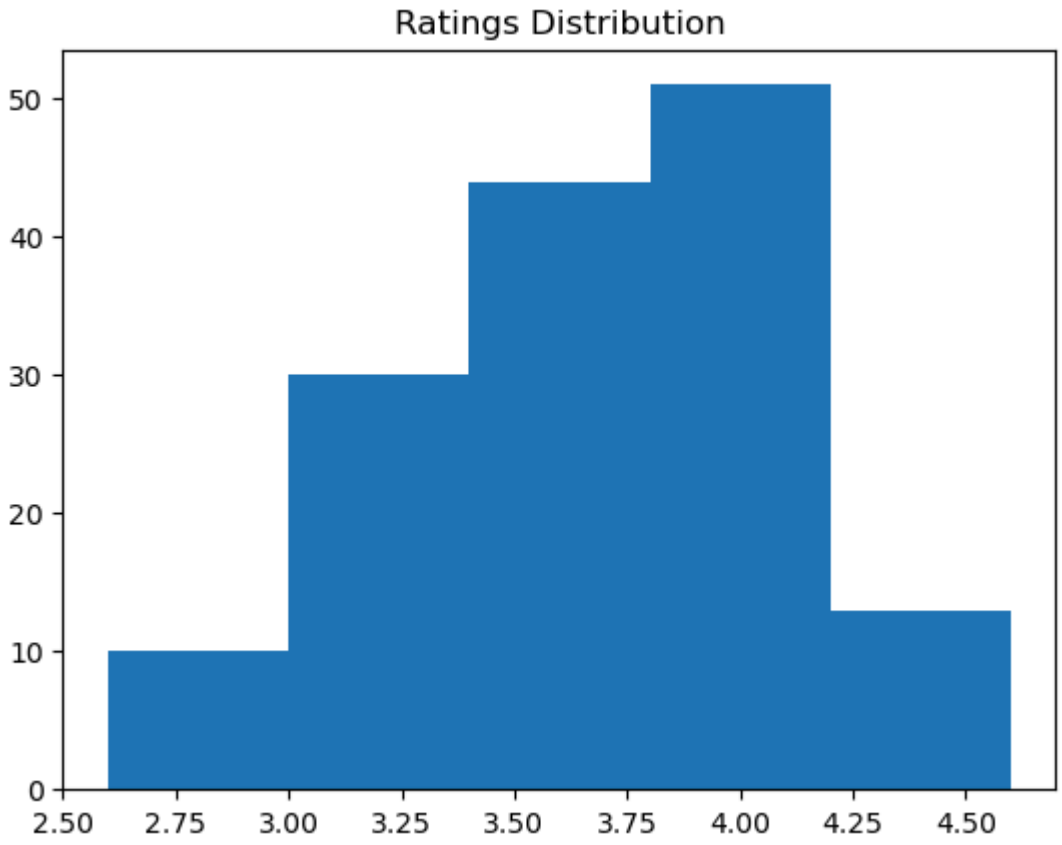
Out[155…

```
<Axes: xlabel='online_order', ylabel='count'>
```




In [156…

```
plt.hist(dataframe['rate'],bins=5)
plt.title('Ratings Distribution')
plt.show()
```

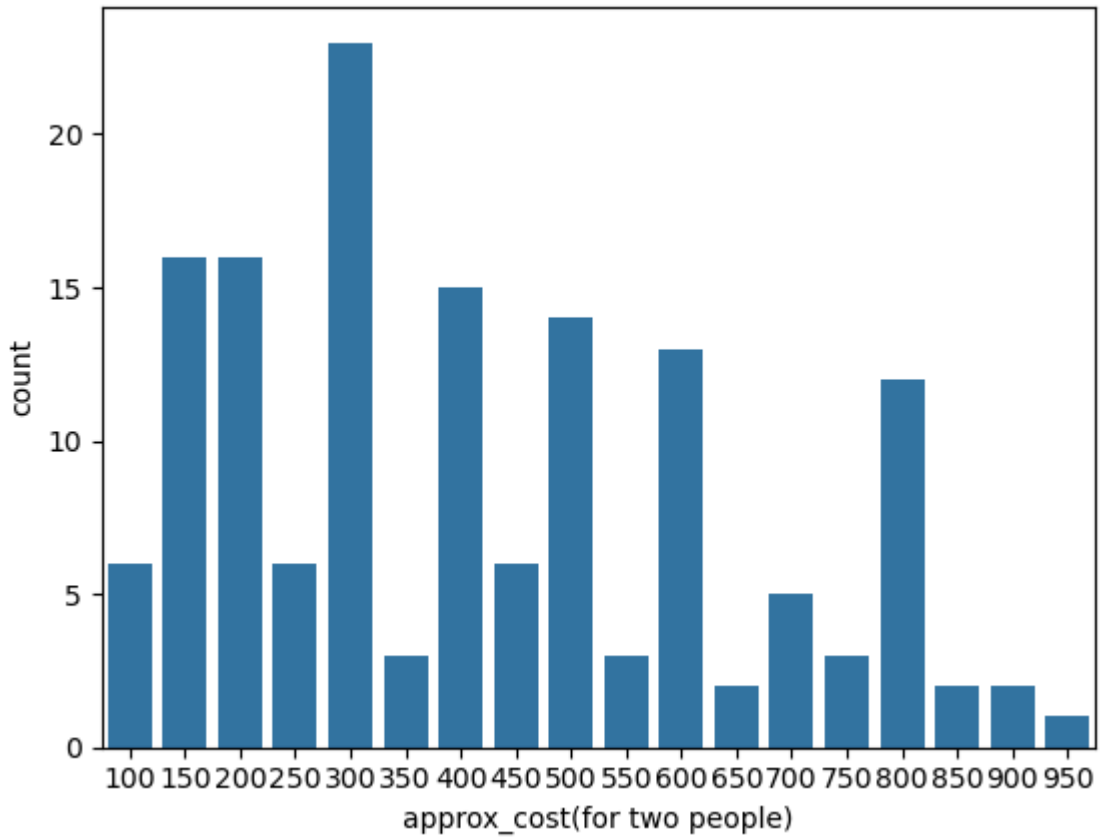



In [157…

```
couple_data=dataframe['approx_cost(for two people)']
sns.countplot(x=couple_data)
```

Out[157…

```
<Axes: xlabel='approx_cost(for two people)', ylabel='count'>
```

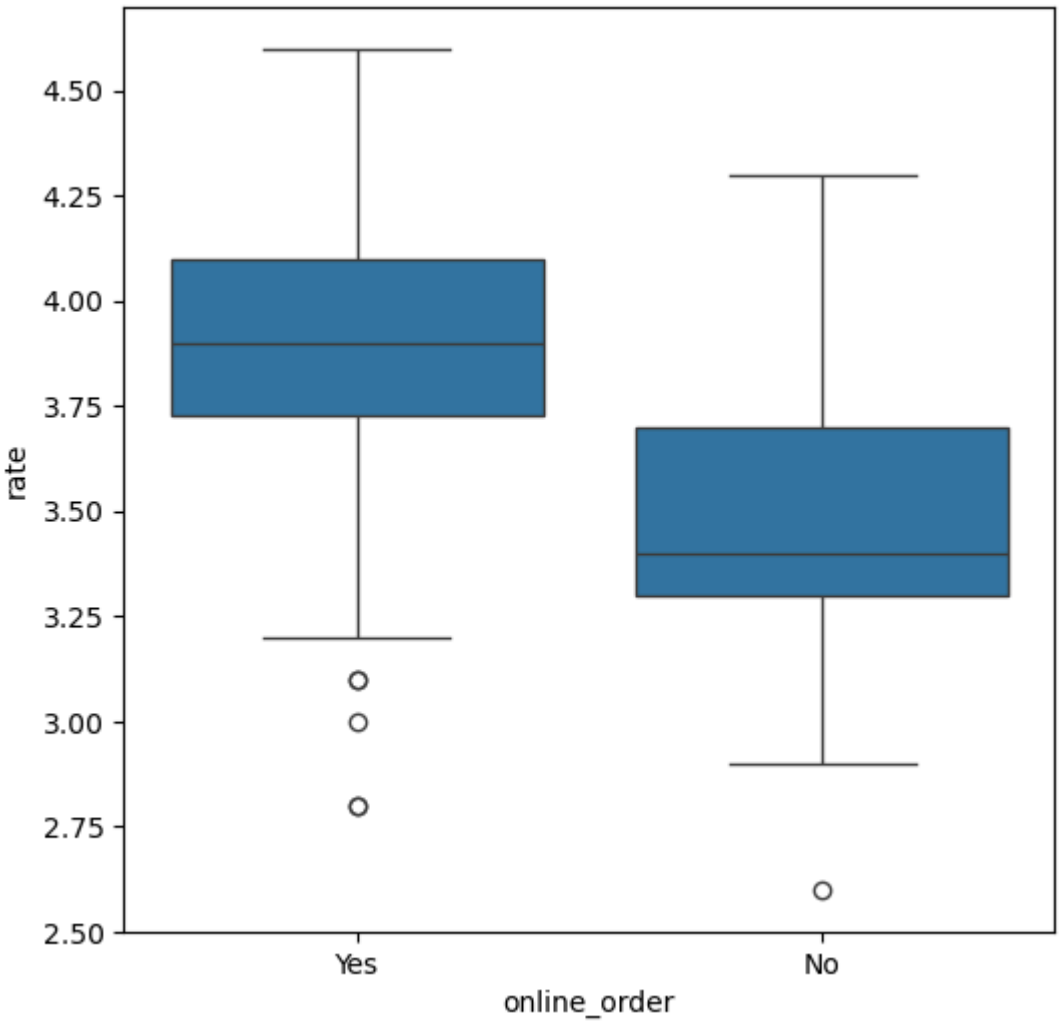



In [158…

```
plt.figure(figsize = (6,6))
sns.boxplot(x = 'online_order', y = 'rate', data = dataframe)
```

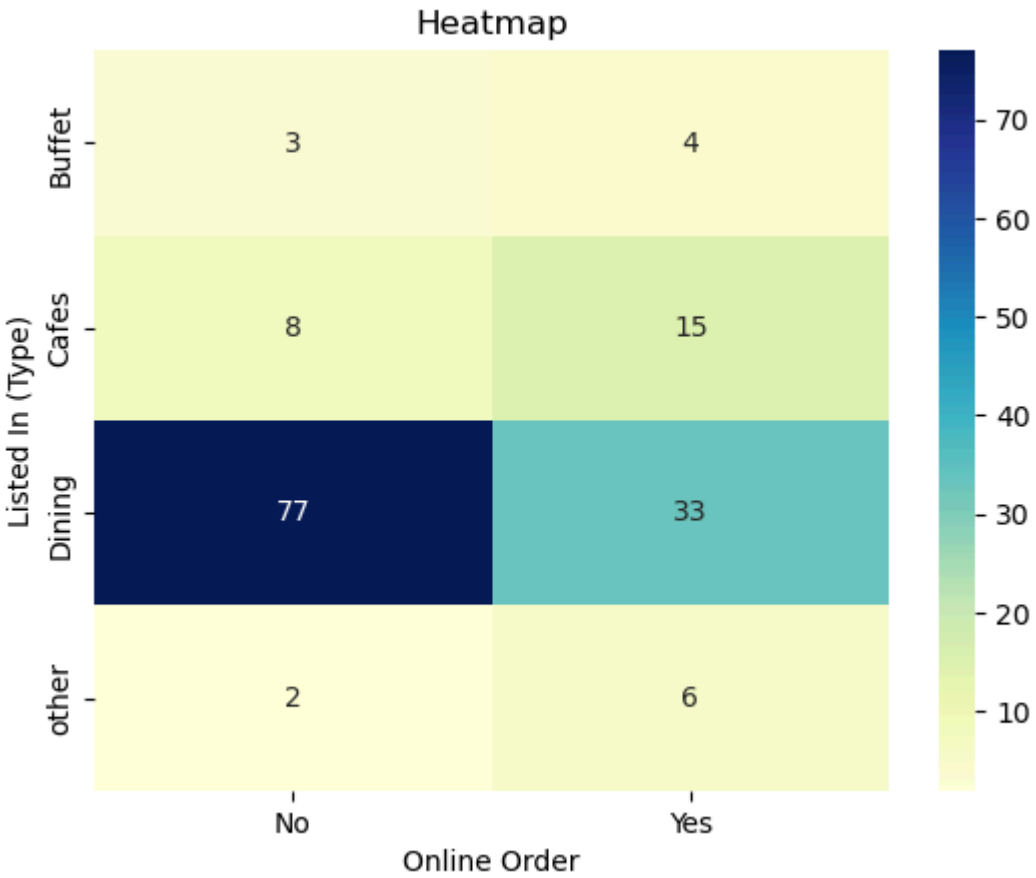
Out[158…

```
<Axes: xlabel='online_order', ylabel='rate'>
```




In [159…

```
pivot_table = dataframe.pivot_table(index='listed_in(type)', columns='online_order', aggfunc='size', fill_value=0)
sns.heatmap(pivot_table, annot=True, cmap='YlGnBu', fmt='d')
plt.title('Heatmap')
plt.xlabel('Online Order')
plt.ylabel('Listed In (Type)')
plt.show()
```


Heatmap
Buffet
Cafes
Dining
other
Listed In (Type)
3
4
8
15
77
33
2
6
No
Yes
Online Order
70
60
50
40
30
20
10

In [ ]:

In [ ]: